UNIT 4

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What is a Graph?

- **Graph** is a non-linear data structure consisting of vertices and edges.
- The vertices are sometimes also referred to as nodes and the edges are lines or arcs that connect any two nodes in the graph.
- More formally a Graph is composed of a set of vertices(**V**) and a set of edges(**E**). The graph is denoted by **G(V, E).**

Components of Graph

- Vertices
- Edges

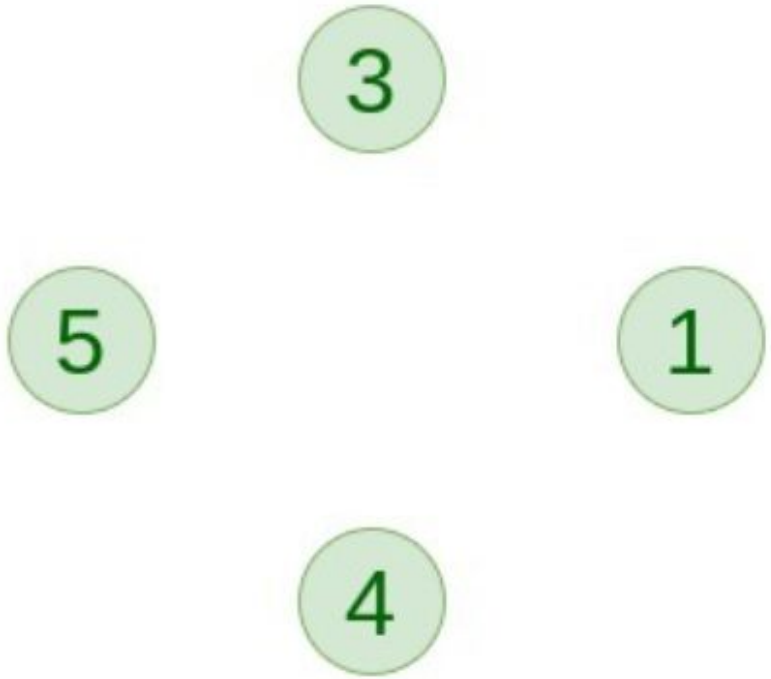
Types of Graph

1.Null Graph

A graph is known as a null graph if there are no edges in the graph.

2. Trivial Graph

- Graph having only a single vertex, it is also the smallest graph possible.



Null Graph

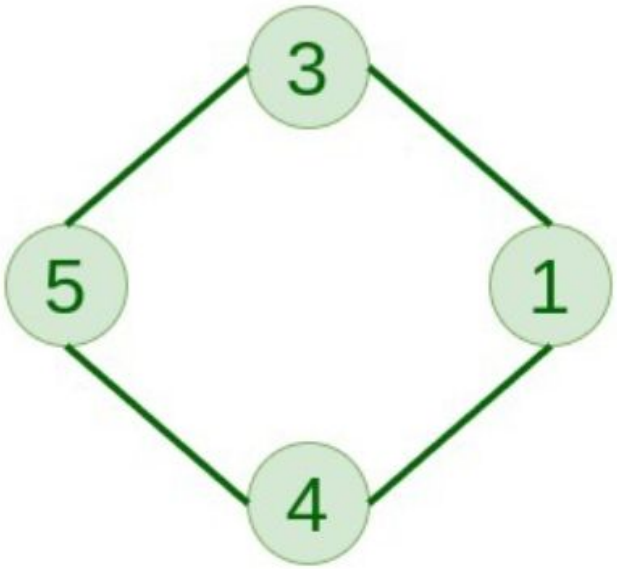
Trivial Graph

3. Undirected Graph

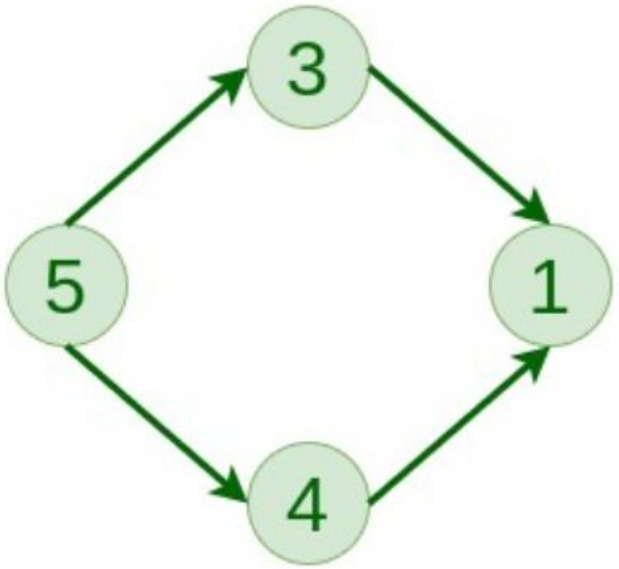
- A graph in which edges do not have any direction. That is the nodes are unordered pairs in the definition of every edge.

4. Directed Graph

- A graph in which edge has direction. That is the nodes are ordered pairs in the definition of every edge.



Undirected Graph



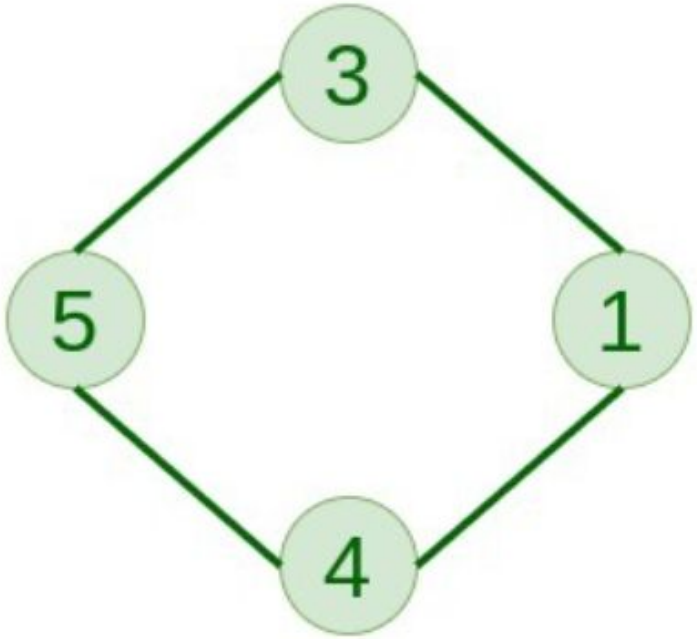
Directed Graph

5. Connected Graph

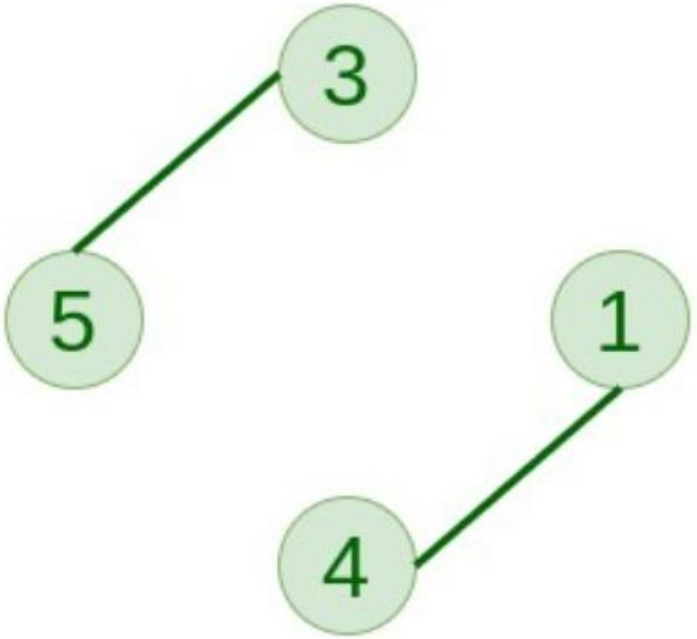
The graph in which from one node we can visit any other node in the graph is known as a connected graph.

6. Disconnected Graph

- The graph in which at least one node is not reachable from a node is known as a disconnected graph.



Connected Graph



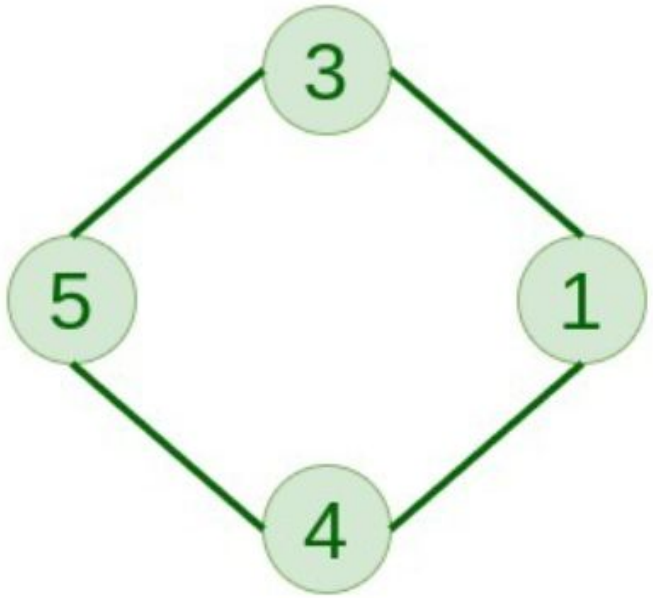
Disconnected Graph

7. Regular Graph

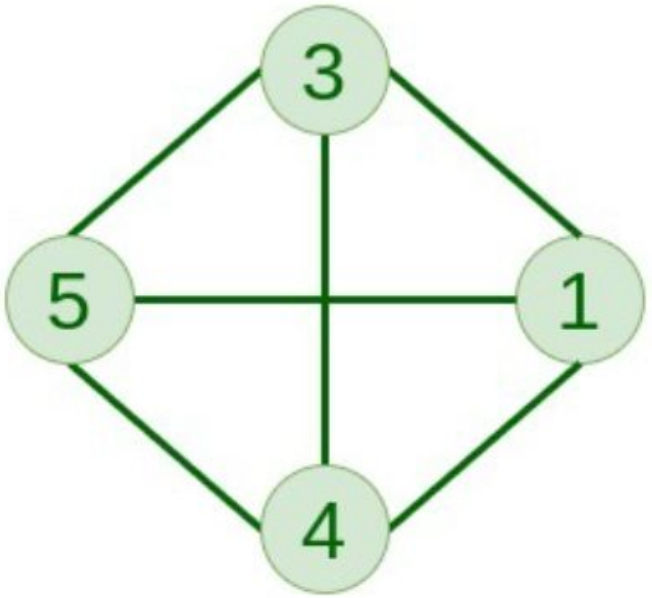
The graph in which the degree of every vertex is equal to K is called K regular graph.

8. Complete Graph

- The graph in which from each node there is an edge to each other node.



2-Regular



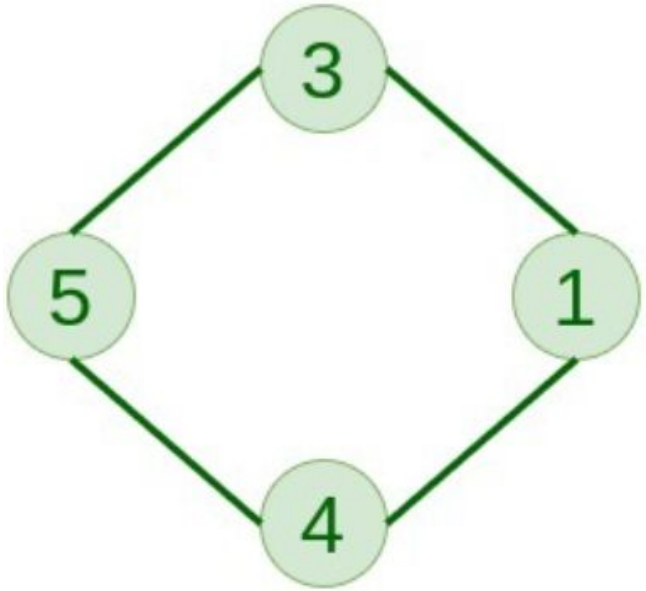
Complete Graph

9. Cycle Graph

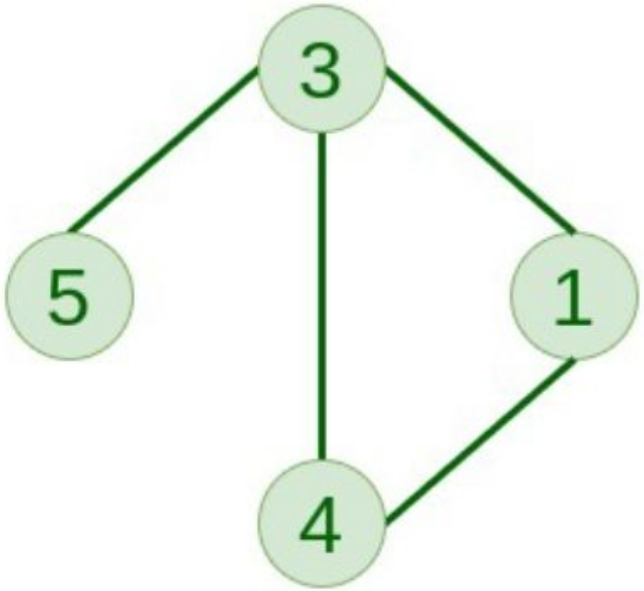
The graph in which the graph is a cycle in itself, the minimum value of degree of each vertex is 2.

10. Cyclic Graph

- A graph containing at least one cycle is known as a Cyclic graph.



Cycle Graph



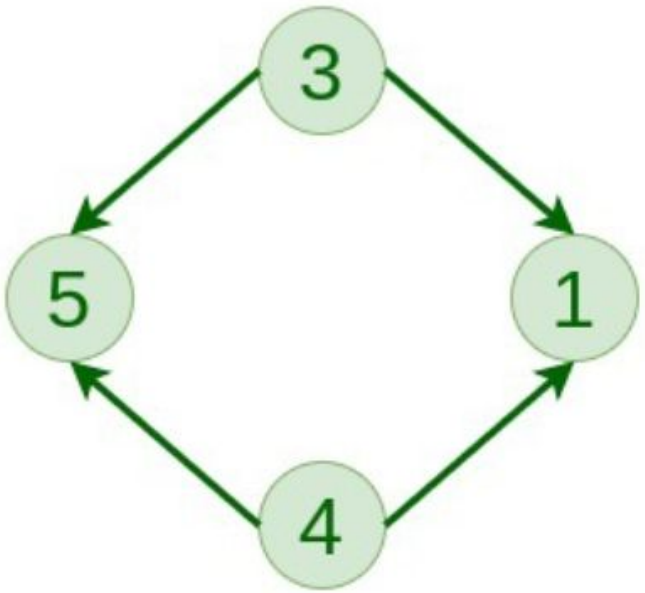
Cyclic Graph

11. Directed Acyclic Graph

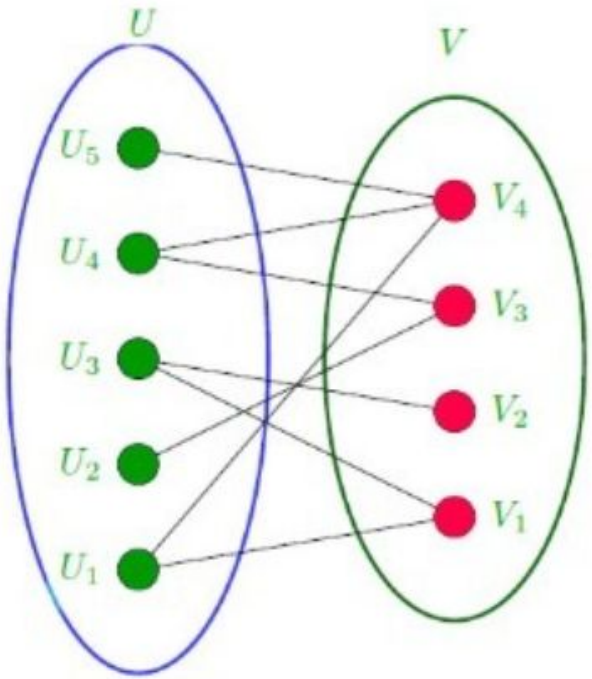
- A Directed Graph that does not contain any cycle.

12. Bipartite Graph

- A graph in which vertex can be divided into two sets such that vertex in each set does not contain any edge between them.



Directed Acyclic Graph



Bipartite Graph

13. Weighted Graph

- A graph in which the edges are already specified with suitable weight is known as a weighted graph.
- Weighted graphs can be further classified as directed weighted graphs and undirected weighted graphs.

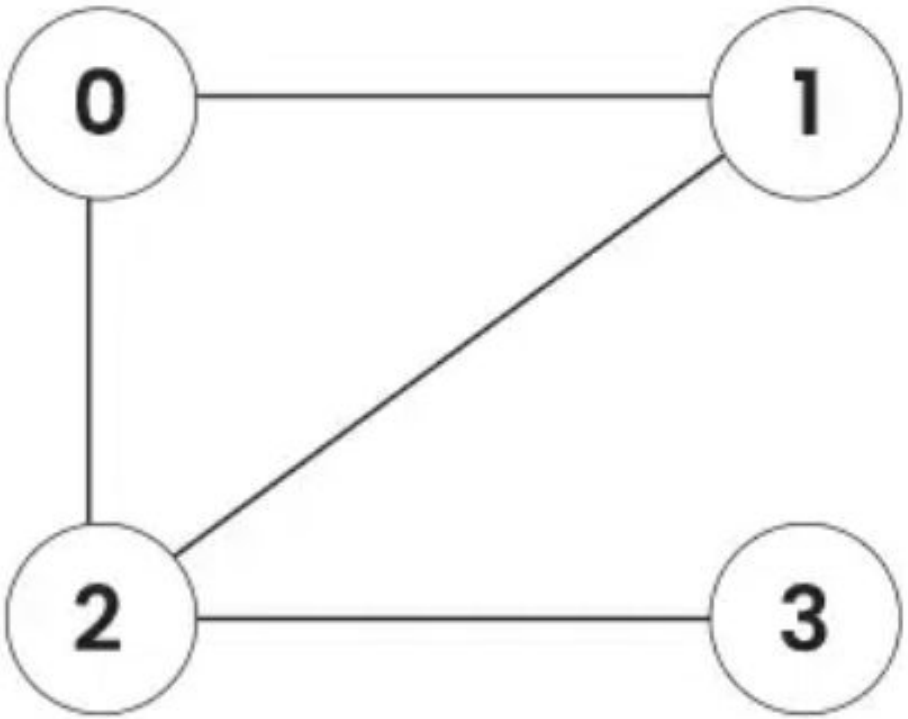
Representation of Graph

- Adjacency Matrix
- Adjacency List
- Incidence Matrix

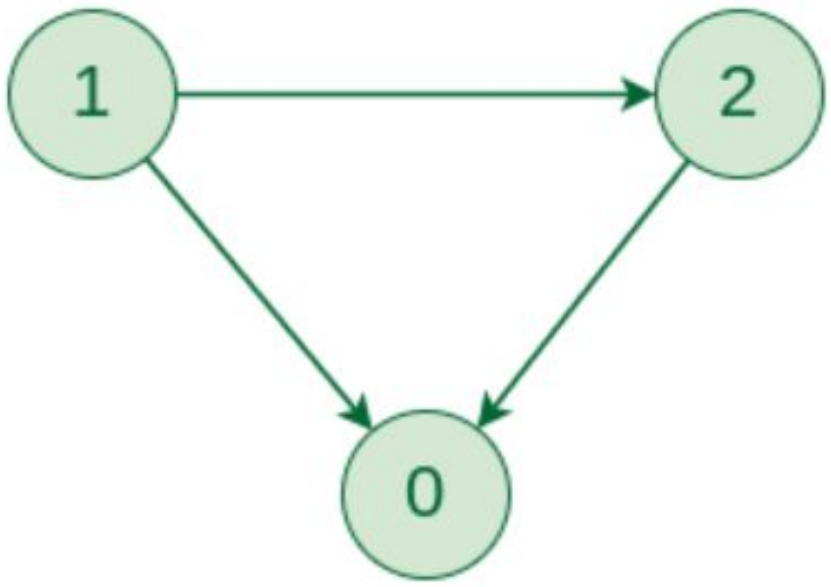
Adjacency Matrix

- In this method, the graph is stored in the form of the 2D matrix where rows and columns denote vertices.
- Each entry in the matrix represents the weight of the edge between those vertices.

Adjacency Matrix of Graph



	0	1	2	3
0	0	1	1	0
1	1	0	1	0
2	1	1	0	1
3	0	0	1	0



Directed Graph

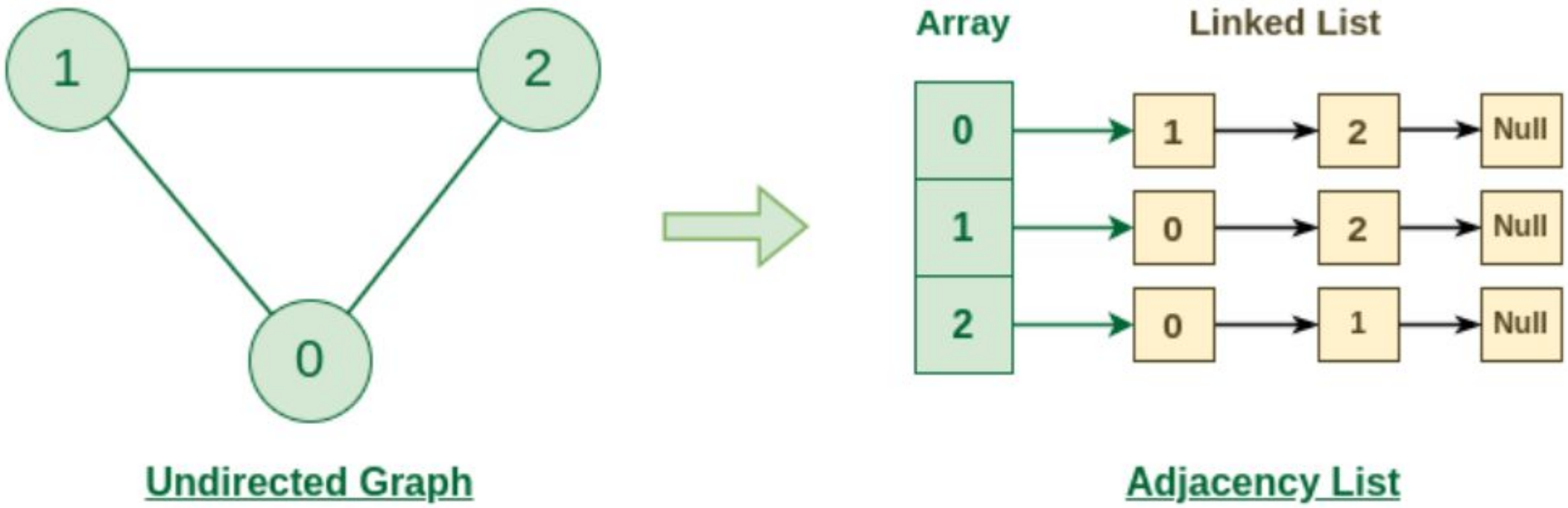
	0	1	2
0			
1	1		1
2	1		

Adjacency Matrix

Graph Representation of Directed graph to Adjacency Matrix

Adjacency List

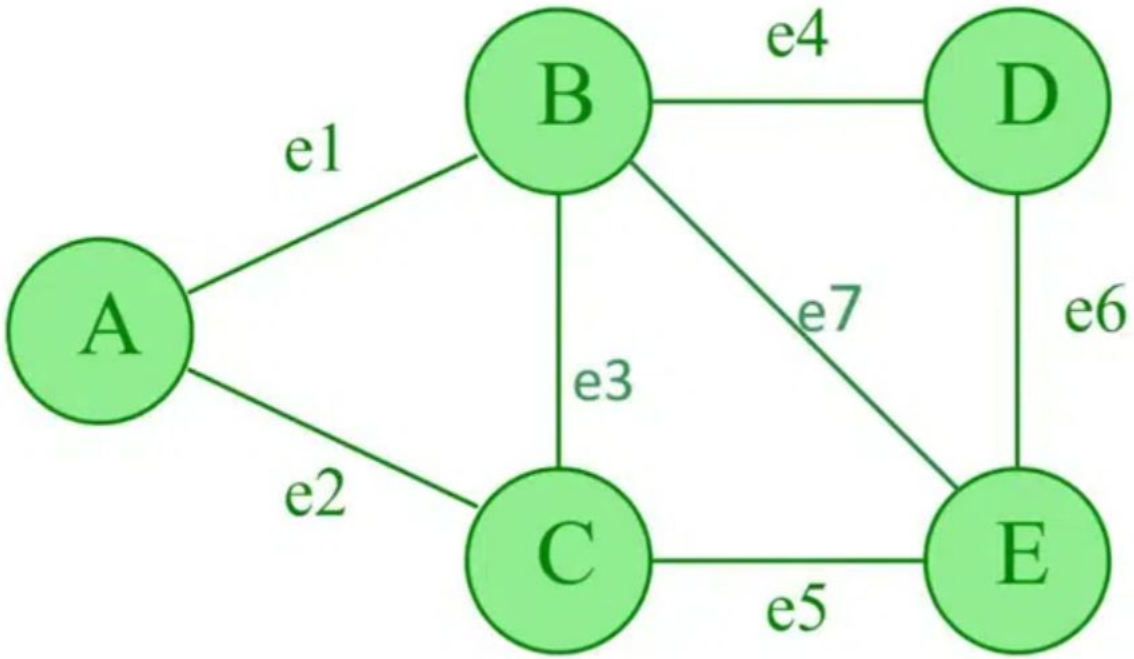
- An adjacency list is a data structure used to represent a graph where each node in the graph stores a list of its neighboring vertices.



Graph Representation of Undirected graph to Adjacency List

Incidence Matrix

- The **incidence matrix** is a way of representing a graph where rows represents vertices and columns represent edges, each entry in the matrix indicates whether a vertex is incident to an edge.



Node/Edges	e1	e2	e3	e4	e5	e6	e7
A	1	1	0	0	0	0	0
B	1	0	1	1	0	0	1
C	0	1	1	0	1	0	0
D	0	0	0	1	0	1	0
E	0	0	0	0	1	1	1